

Department of Mathematics and Applied Mathematics

3DM: Discrete Mathematics

Exam

1200 Wednesday 30 September – 1200 Thursday 1 October 2020

50 marks = 100%



- This question paper consists of **11 pages** (including this one).
- You must upload your solution to the Assignments Tool in the 3DM Vula no later than 12 noon Thursday 1 October.
- Please put your name and student number on your solution.
- In Question 1, you must just give the answer (i.e., no proofs/working necessary). In Question 2, you must show all working/proofs (no working/proof = no marks, even if your answer is correct).
- You may type your solution (I strongly suggest if you do that you use $\text{\LaTeX}$) or (if you have a tablet) write on this question paper and upload the PDF.
- Or you may write your solution out and upload photographs/scans of the pages. If you do this, please make sure each page is clearly legible. **If the photograph is not clear or if we can't read your handwriting, we will mark your solution incorrect.**
- The mathematics in this exam should only take 2 – 3 hours to finish. The rest of the time is to accommodate difficulties connecting etc. You should therefore finish the test as soon as possible and upload your solution as soon as you are finished. **Don't wait till the last minute to upload your solution!**

OPEN BOOK TEST RULES

This is an Open Book test. That means:

- You may refer to the course notes and your homework answers.
- You may not use the internet to get answers or help. I had to think about this one for a while. Not everyone in the class can easily access the internet. Allowing students to cruise around the internet looking for things to help them with a test they're busy writing risks creating a very uneven playing field.
- You may not communicate with anyone about the questions (that includes other students in the course, people on the internet, your grandmother the maths professor, your friend who did the course 3 years ago, etc).

Please read the Faculty of Science Student Honour Code, below.

FACULTY OF SCIENCE STUDENT HONOUR CODE

The University of Cape Town's official statement of general rules and policies, including a statement of values and expected student and staff conduct, can be found [here](#). All staff and students in the Faculty of Science are required to familiarise themselves with UCT's rules and policies. The Faculty of Science takes UCT's statement of values seriously. Among other things, every member of the Faculty of Science — student and staff — is expected to strive for personal and academic integrity; to be respectful and tolerant towards others; to honour the rights, personal choices and property of others; to take responsibility for their behaviour individually and in groups; and generally to act as a responsible citizen in our academic community and beyond. As stated in the above mentioned document: "Truth, fairness, consistency and integrity in both academic and other work, and in all personal and institutional relationships" are key elements of UCT's academic values. We are currently in a global crisis, and recognise that distance learning will be a challenge for many students. Nonetheless, we must strive to uphold these values. We ask students to commit to an Honour Code — i.e. doing what is morally right — when completing tasks remotely. In particular, all assignments and tests must be a student's own work, with no input from others (unless otherwise specified), and no work can be plagiarised. We also expect students to report to the Faculty if they believe that other UCT students have breached this code. The Faculty of Science supports our students in this difficult time, and requests their full commitment to the integrity of the academic project. Together we will persevere, and succeed.

Question 1. [15 points] Give an example of each of the following (if such a graph exists) or write Does Not Exist (if it doesn't). **Do not show working!**

(i) A nontrivial graph in which no two vertices have the same degree . (3)

(ii) A tree of radius 5 and diameter 7. (3)

(iii) A forest that is not bipartite. (3)

(iv) A connected graph that has exactly two cut-vertices but no bridge. (3)

(v) A bipartite graph G of order 10 such that (i) for every pair u, v of nonadjacent vertices, the graph obtained from G by adding an edge between u and v is not bipartite, and, (ii) the graph G does not contain a subgraph isomorphic to $K_{2,3}$. (3)

Question 2. [35 points] **Show all working/proofs!**

- a. Prove or disprove: There exists a graph with center $K_3 + \overline{P_2}$. (5)

- b. If G is a graph of order $n \geq 3$ with $\delta(G) \geq \frac{n}{2}$, prove that G is a block. (5)

- c. Let G be a connected graph. Prove that if k is some integer satisfying $\text{rad } G < k \leq \text{diam } G$, then G contains (at least) two vertices of eccentricity k . You may assume, without proof, the following Lemma: *If c is a central vertex of G and u' an eccentric vertex of u and $e(u) > \text{rad } G$, then u does not lie between c and u' .* (5)

d. Suppose that G is a connected graph of order 3 or more. Let $c(G)$ denote the number of cut-vertices of G and $b(G)$ denote the number of bridges of G .

(i) Prove or disprove: $c(G) \leq b(G)$. (5)

(ii) Prove or disprove: $b(G) \leq c(G)$.

(5)

- e. Let G be a nontrivial bipartite graph. If $B = \{V_1, V_2\}$ is a bipartition of G , we define the *bipartition graph* G_B to be the graph with vertex set

$$V(G_B) = V(G)$$

and edge set

$$E(G_B) = \{uv : u \text{ and } v \text{ are in the same partite set of } B\},$$

i.e., u and v are adjacent in G_B if both u and v are in V_1 , or both u and v are in V_2 . For example, if $G = C_6 : v_1, v_2, v_3, v_4, v_5, v_6, v_1$ and $B = \{\{v_1, v_3, v_5\}, \{v_2, v_4, v_6\}\}$, then $G_B \cong 2C_3$.

- (i) Prove or disprove: If B and B' are bipartitions of a nontrivial bipartite graph G , then $G_B \cong G_{B'}$. (5)

- (ii) Prove or disprove: If B and B' are bipartitions of a nontrivial connected bipartite graph G , then $G_B \cong G_{B'}$. (5)